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Being Human in the Age of Generative AI: Young People's Ethical Concerns about Writing and Living with Machines

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ABSTRACT

The recent unveiling of chatbots such as ChatGPT has catalyzed vigorous debates about generative AI's impact on how learners read, write, and communicate. Largely missing from these debates is careful consideration of how young people are experiencing AI in their everyday lives and how they are making sense of the questions that these rapidly evolving cultural tools raise about ethics, power, and social participation. Engaging cultural-historical perspectives on technology, the present study drew on student survey and focus group data from English language arts classes in two culturally and linguistically diverse high schools to answer the following questions: (1) How are young people using AI in their everyday lives, if at all?; (2) What do young people identify as key considerations related to AI-mediated writing?; and (3) What ethical and critical considerations, if any, inform young people's sensemaking of and practices with AI? Young people reported using generative AI for diverse purposes in and out of school, including to accomplish routine organizational and information tasks, to entertain themselves through experimenting with AI technologies, and to catalyze their thinking and writing processes. Survey and focus group participants' responses suggested their regular navigation of ethical and critical dimensions of AI use and their contemplation of what it means to be human through and with advancing technologies. Young people also reported a lack of opportunity to examine AI practices and perspectives in school, suggesting the important role schools can play in supporting youths' development of AI ethics.

While artificial intelligence has been used in education for over six decades, recent advancements in AI technology have provoked renewed attention to these tools and the challenges and opportunities they present (Buolamwini, 2023; Streitfeld, 2023). The recent public unveiling of large language models such as ChatGPT—and their ensuing integration into commercial platforms—has catalyzed vigorous debates about generative AI's impact on how learners read, write, and communicate. Practitioners, researchers, policymakers, media outlets, and AI developers have contributed to the growing discourse around the benefits and risks of AI in education, with some voicing hope for educational transformation and others sharing concerns related to epistemic shifts, pedagogy, and ethical dilemmas.

Less visible in these deliberations of generative AI are the voices and perspectives of young people. While ample research suggests that young people are not merely skillful consumers of technology but also agentive creators and critics (Gutiérrez et al., 2019; Mirra et al., 2018; Stornaiuolo & Thomas, 2018; Vakil & McKinney de Royston, 2022), a construction of complacent youth eager to leverage AI to "cheat" on coursework

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dominates the public discourse and imagination (Spector, 2023; Vaidhyanathan, 2023). Such a restricted understanding of young people's stances toward and practices with AI fails to capture the complexities of how they are making sense of and composing with generative AI in their everyday lives, an oversight with increasingly urgent implications as the most-used commercial composing platforms (e.g., Microsoft Word, Google Docs) race to embed AI tools in their interfaces. Emerging scholarship offers an initial glimpse into how young people understand the tensions and dilemmas they face as generative AI complicates notions of "reading" and "writing" (e.g., Malmström et al., 2023; Morales-Navarro et al., 2023; Singer, 2023), but young people's voices are still absent from many conversations about AI and writing (e.g., Tate et al., 2023), even as some scholars begin to think with youth about AI and media (e.g., Lee et al., 2022). Young people, particularly those from nondominant communities, are often left out of conversations that directly impact their futures. With AI's potential to perpetuate discrimination based on multiple identities (Buolamwini, 2023), how young people who experience various forms of marginalization negotiate these technologies in everyday life has profound implications for social and civic participation. To ensure decisions and policies related to AI and education reflect the realities of young people, their voices must be at the forefront of discussions and decision-making. Relatedly, as the field of literacies studies develops a critical research agenda around AI, centering young people's perspectives is imperative to the design of conceptual and empirical endeavors that ground evolving literacies in socially just commitments (e.g., Lyiscott et al., 2020; Mirra et al., 2018).

We center young people's everyday experiences with and perspectives on AI-mediated writing in this qualitative study, conducted with high school students who participated in a yearlong social design-based experiment (Gutiérrez, 2016) that explored secondary English language arts (ELA) classes as sites for interdisciplinary critical computing education (Vakil & Higgs, 2019). We draw on student survey and focus group data from ELA classes in two culturally and linguistically diverse high schools to explore the following questions:

1. How are young people using AI in their everyday lives, if at all?
2. What do young people identify as key considerations related to AI-mediated writing?
3. What ethical and critical considerations, if any, inform young people's sense- making of and practices with AI?

We found that ethical considerations are a major factor for youth when composing with AI and that they are grappling with how to take critical, principled approaches, often with little guidance from school. One of their core concerns centers on the link between writing and what it means to be human—specifically, how writers' authenticity and creativity are threatened when the labor of meaning-making can be delegated to machines. Our participants' reflections on the nature of writing, the erosion of personal voice, and the ownership of ideas suggest their uncertainties about a future in which AI gains an ever-stronger presence. In short, we found that young people are thinking deeply about these new technologies in their everyday composing practices.

As these technologies become more intertwined with commercial platforms and composing tools, educators will play a vital role in supporting students in navigating new and changing composing landscapes. Yet current conversations about AI in schools may not include the complexity of young people's writing practices and the ethical and aesthetic concerns about writing and humanity that they grapple with. Through this research, we hope to help change the conversation around AI so that youth perspectives are central to understanding implications of AI in local and global contexts, and students' and teachers' ethical and aesthetic choices are acknowledged as vital components of teaching and learning in the age of generative AI.

Theoretical Framework

Our study is designed around the premise that young people are "philosophers of technology," an orientation toward youth as actors who take up moral and ethical considerations of technology "to catalyze social, cultural, or political transformation" (Vakil & McKinney de Royston, 2022, p. 4). Vakil and McKinney de Royston propose that young people can become philosophers of technology through "nuanced contemplation of the ethical complexities and implications of technology" (p. 2). This approach eschews unidimensional or deficit-oriented narratives about young people by emphasizing their agency as "historical actors" who can work toward social transformation by interrogating their everyday practices, including their literacy practices (Gutiérrez et al., 2019, p. 291). Positioning youth as philosophers highlights the importance of contemplation "as a privileged mode of sensemaking with and about technology that invites in aesthetics and centers ethics (Schiller, 2016)" (Vakil & McKinney de Royston, 2022, p. 340). With this framing, we recognize that youth already contemplate the ethical, critical, and aesthetic dimensions of technology in their everyday experiences, even as we acknowledge also the importance of opportunities for young people to engage in philosophical sensemaking and critical inquiry in more formalized learning contexts such as classrooms.

By viewing young people as philosophers of technology, we aimed to avoid simplistic labels (e.g., technology as

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good/bad; youth as consumers/producers) and acknowledge the multiplicity, relationality, and contingency involved as emerging digital technologies are imbricated in everyday social, cultural, economic, and political activity. From this perspective, any technology must be understood culturally and historically, not as an isolated tool or machine but as a sociotechnical system, or "an entanglement of humans with each other, the tools, the arts, crafts, practices, and systematized, institutionalized ways of life of their time and circumstances" (Pea & Cole, 2019, p. 16). Within a sociotechnical system, a tool is an important element—but still only an element—of a convergence of conditions that result in particular practices with particular consequences for actors in that system. These sociocultural practices matter, both as instantiations of these systems in everyday life and as mediators of interactions that can reorganize social life. A fundamental understanding of a technology from this perspective is that a technology entails the doing of something and the theorizing of that doing (Pea & Cole, 2019). A cultural-historical lens reminds us that all technologies, whether as ancient as written language or as contemporary as AI-mediated writing, are shaped by and reflect the social and cultural contexts of their creation and use. This approach aligns with our view of the young people in our study as philosophers of technology, who are actively engaging technologies and contemplating "the enduring questions about what technology is, what it should or could be and do, and assess the moral complexities" (Vakil & McKinney de Royston, 2022, p. 340) in relation to broader social, cultural, political, and economic forces.

Views of AI as a sociotechnical system can be put in conversation with recent literacy scholarship that explores questions about how people live and learn with these machines from a posthuman perspective (Leander & Burriss, 2020). Posthuman approaches in literacy studies focus on the instability of meanings as they are distributed across material networks of human and non-human actors, a decentering of traditional notions of human rationality and transparency (Nichols & Campano, 2017) and a centering of affect, emergence, and indeterminacy (Ehret & Rowsell, 2021; Kuby et al., 2019). While such perspectives push the field to think more deeply about what it means to be a human engaging in literate activity amidst these emergent and precarious ecological relations, we contend that putting them in conversation with cultural-historical approaches can enrich understandings of composing with AI by highlighting the historical and situated nature of distributed knowledge creation. Grounding our study in an approach to technology as both a "practice and a theory of that practice" (Pea & Cole, 2019, p. 15), we are interested in the multiple, sociohistorical dimensions "entangled" in the key human technology of writing: people, histories, material and psychological tools, sociopolitical dynamics, skills, knowledge, bodily coordination, and spatial and temporal circumstances. Such a focus emphasizes that writing with AI is more than just "doing" writing; rather, it is always a situated activity mediated by human and non-human relations that encompasses our interactions with and interpretations of technologies. These overlapping lenses suggest that writing with AI is as much about the act of mediated composing (i.e., the doing) as it is about contemplation, or questioning, pondering, resisting, and otherwise making sense of what it means to live in evolving sociotechnical systems (i.e., theorizing about the doing).

Our study centers on these sensemaking capacities of young people as they take up emerging technologies to write. We are particularly alert to the ethical dimensions of youths' sensemaking with new technologies, which literacy researchers have explored in relation to youths' multimodal, cosmopolitan literacy practices (e.g., Hull & Stornaiuolo, 2014; Sun, 2023; Vasudevan, 2014). As young people engage in AI-mediated composition and contemplate those composing practices, they are grappling with ethical, critical, and aesthetic dimensions of communication in the context of globalization that put under scrutiny their responsibilities toward and relations with "others" in a mediated space (Silverstone, 2007). "Others" here include not just other people but places, technologies, and histories that are involved in an "ethics of relationality"; these concerns are also at the heart of an "ethics of technology" that youth must contemplate as philosophers of technology (Vakil & McKinney de Royston, 2022, p. 340). While AI ethics should be of concern to anyone who encounters AI in their day-to-day life (in other words, all of us), young people are often the most precariously positioned within quickly changing sociotechnical systems, particularly youth from nondominant communities subject to forces of racial and surveillance capitalism intertwined with new writing technologies (e.g., de Roock, 2021; Dixon-Román et al., 2020).

Relevant Literature

Despite widespread use of AI in everyday technologies (e.g., music recommendations, email assistance), its potential in education has only recently gained widespread attention. Notably, AI writing tools have been used for decades (Warschauer & Ware, 2006). Since the 1960s, machine learning systems have been developed to provide automated writing evaluation and revision assistance to student writers, with recent advances in AI offering a new wave of development that includes intelligent tutoring systems, text generation, and personalized feedback (Cotos, 2023). While research in this area provides valuable context for thinking about writing instruction with machines, we found it less applicable to our study of youth experiences and understandings of AI, particularly in light of this area's tendency toward narrower conceptualizations

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of writing, platforms, and equity (see Stornaiuolo et al., 2023). Instead, we consider here how prior research exploring the relationships among youth, writing, and emerging digital technologies offers important insights for understanding how youth in our study were using and contemplating generative AI.

As digital technologies have accelerated what Brandt (2014) called a societal shift toward mass writing in the 21st century, it is perhaps unsurprising that young people are leaders in exploring these technologies in their everyday writing lives. The turn toward generative AI represents the latest addition to the ever-expanding array of technologies encountered by young people in their digital writing practices, as they compose, remix, orchestrate, and assemble symbolic artifacts across multiple modes, media, and languages using digital technologies (Freedman et al., 2016; Stornaiuolo, 2015). Literacy researchers have long explored how youth compose with emerging technologies outside of school contexts based on their interests, passions, and identities (e.g., Aguilera, 2021; Chandler-Olcott & Mahar, 2003; Curwood et al., 2013; Hull & Katz, 2006). This research highlights the interplay between ethical and aesthetic considerations within these new forms of digital writing in globally connected environments (e.g., Aragon & Davis, 2019; Hull & Nelson, 2009; Hull & Stornaiuolo, 2014).

Building on this foundational work, scholars have turned toward theorizing young people's digital writing in and across school and non-school contexts (e.g., Hicks, 2018; Lammers & Marsh, 2015; Mills & Exley, 2014; Pandya & Sefton-Green, 2021). Pandya and Sefton-Green (2021) argue that digital writing now includes "all forms of communication, expression and creativity taking place through, on or with digital technologies and digital platforms" (pp. 113–114). Defining digital writing in this way highlights the role of platforms, which scholars in critical platform studies in education (Nichols & Garcia, 2022) have shown to exert powerful influence over youths' literacy practices (e.g., LeBlanc et al., 2023; Robinson, 2022a, 2022b). As de Roock (2021) explains, "digital writing is increasingly mediated through commercial digital apps and other interactive platforms. Algorithms thus play the role of coauthors with profit motives" (p. 187). These platforms on and through which people compose, such as commercial AI platforms like ChatGPT, conceal their profit goals and data practices while seeming like neutral intermediaries (Pangrazio et al., 2022). The present lack of transparency regarding the role of platforms and algorithms in digital writing is likely to intensify as AI technologies become deeply embedded in composing tools and these features become even less visible to users during the composing experience.

While literacy researchers have begun to examine how algorithms, platforms, and interfaces can explicitly shape youths' digital writing practices (e.g., Monea, 2020; Robinson, 2022b), several studies have explored the role of AI technologies specifically. Dixon-Román et al. (2020), for example, drew on new materialist and Black feminist thought to investigate how one AI platform (Essay Helper) offering automated writing feedback inscribed racialized logics that narrowed and constrained possibilities for writing. Following this critical approach to thinking about AI and writing, de Roock (2021) and Robinson (2022a) both interrogated how automation positions digital writers in relation to racial capitalism and "enrolls and entangles the writer in emerging forms of ideological, political, extractive labor arrangements" (de Roock, 2021, p. 2). Echoing concerns voiced by young people in our study, Robinson (2022a) suggested that new forms of generative AI like ChatGPT-3 will influence writers' subjectivity—who they *become* as writers—and how they come to know themselves, others, and the world. Finally, Lee et al. (2022) developed an ethics-centered pedagogy to help youth journalists and artists in their out-of-school program learn about AI as they composed with AI technologies. They found that ethically designed pedagogies supported youth composers in the development of agency in relation to AI technologies, as youth opted out of being complicit with the technologies and resisted the erosion of self that they experienced when using AI technologies. These insights hold promise for what ethics-oriented pedagogies might look like in school contexts, an implication we later discuss in relation to young people in our study.

While the substantial and still-growing body of digital writing research noted above is reflective of the education research community's broader interest in the role of new media in society, scholarship on technology-related learning has often centered on technology as an aid rather than the object of learning itself (Vakil & McKinney de Royston, 2022). Attuned to recent work that interrogates biases embedded in the computing technologies that shape our lives (e.g., Benjamin, 2019; Buolamwini & Gebru, 2018; Noble, 2018), education researchers in recent years have advocated for learning opportunities that go beyond uses of tools to learning that explicitly situates tool use within particular systems. Specifically, they have asked critical questions related to power, ethics, and social impact, and how students learn or do not learn about the complex social, symbolic, and material relationships embedded in the tools that they use in their everyday lives (e.g., Kafai & Proctor, 2022; Nichols & Garcia, 2022; Philip et al., 2016; Vakil, 2018; Vakil & Higgs, 2019). These scholars posit that technology-related education should aim to help students develop capacity to interrogate and situate technologies within broader sociopolitical systems in tandem with their burgeoning technical skills.

While opportunities to pursue learning *about* technologies have been less common outside of STEM learning contexts, recent work in the fields of English and social studies education indicates growing interest in digital

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technologies as objects of study. Scholars in these disciplines have begun to examine the social, material, and ethical complexities that are embedded in and performed by the technologies used in everyday life (e.g., Garcia & Nichols, 2021; Higgs, 2020; Krutka et al., 2019; Lizárraga, 2023; Price-Dennis & Carrion, 2017). This marks a shift in focus for sociocultural literacy studies, which historically have focused on the varied literacies mediated by digital technologies or on what it means to participate in a particular type of literacy in the digital age (Freedman et al., 2016; Mirra & Garcia, 2021). For scholars of writing, this turn has highlighted a need to blend ecological approaches to writing and ecological approaches to technologies (Stornaiuolo et al., 2023).

Methods

This study of young people's perspectives on and experiences with generative AI was conducted with students who had participated in Jenni's social design-based experiment (SDBE; Gutiérrez, 2016) that explored English language arts classes as learning sites of sociotechnical awareness, defined as a critical lens that interrogates intersections of computing technologies, power, ethics, and social hierarchies (e.g., race, class, gender). Sociotechnical awareness assumes that technologies are never neutral and, moreover, that technologies can reinforce discrimination and cause harm to marginalized and nondominant communities (e.g., Benjamin, 2019; Eubanks, 2018; Noble, 2018; Tanksley, 2022). A broad goal in developing this lens is to help learners recognize the sociopolitical implications of technologies in the present while also fostering knowledge, skills, and dispositions needed to benefit and transform society in the future.

Although the SDBE is not the focus of this paper, we offer here some details about the design work and the curricula that emerged to contextualize how our participants were selected, as well as resources they may have drawn on in their survey and focus group responses. To support the development of secondary students' sociotechnical awareness, Jenni worked "side by side" (Erickson, 2006) with two teachers, Ms. Kane and Mr. Willis, over the course of an academic year to iteratively design and implement units aimed to develop students' understandings of the social impacts of technology in their English 12 and English 11 classes. The intention was not to "import" sociotechnical awareness as a decontextualized tool, but rather to work together to understand how sociotechnical awareness can be fostered with a variety of meaningful disciplinary texts and practices (e.g., discussion, argumentation, inquiry). For example, students in Ms. Kane's and Mr. Willis' classes read work by critical race and technology scholars Safiya Noble (*Algorithms of Oppression*) and Ruha Benjamin (*Race After Technology*) and explored the "myth" of neutral and objective search engines and how human biases can appear in algorithms. They also participated in ongoing critical examinations of technologies in their own lives, sharing their connections, observations, and questions with classmates, teachers, families, friends, and researchers/report authors through oral presentations, multimodal texts (e.g., posters, infographics, visual representations, slides), essays, digital journals, and reaction letters.

After the launch of ChatGPT in late 2022, Jenni noticed that several students in Ms. Kane's and Mr. Willis' classes mentioned generative AI in the course of class discussions. She had conversations with the teachers to explore how generative AI might be addressed as a topic addressed through the designed curriculum. Although both teachers were interested in learning about ChatGPT and students' experiences with it and other forms of generative AI, time limitations during the school year prevented incorporating AI exploration into the SDBEs. However, Ms. Kane and Mr. Willis supported supplementary data collection proposed by Jenni to learn about their students' positions on and uses of AI for writing. Jenni invited Amy, who shares research interests in digital composition and AI, to think together about how to learn more from participating students about their uses and perspectives related to generative AI.

Context

The study took place at two different high schools during the 2022–2023 school year: Inspiration Academy and Green Hills High School (pseudonyms). Inspiration is a tuition-free public charter school in a socially, culturally, and economically diverse California city. At the time of data collection, Inspiration had 24 full-time teachers and enrolled approximately 350 students in grades 9–12. Approximately 50% of students identified as Latinx, 36% identified as Black, 9% identified as mixed race, 4% identified as white, and under 1% of students identified as Asian, Native American, or Pacific Islander. Approximately 78% of the student population qualified for economic assistance programs.

Green Hills is a large public high school in an affluent, socially and culturally diverse California suburb. At the time of data collection, Green Hills had 115 full-time teachers and enrolled approximately 2800 students in grades 9–12. Approximately 40% of the students identified as Asian, 38% identified as white, 12% as Latinx, 9% as mixed race, 2% as Black, under 1% as Native American or Pacific Islander. Approximately 6% of the student population qualified for economic assistance programs.

Participants

Survey Participants

The survey participants were students in two English 12 and two English 11 classes taught by Ms. Kane at

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Inspiration, and students in three English 12 classes taught by Mr. Willis at Green Hills. All students in these classes who had consented to participate in the IRB-approved SDBE were invited to participate in a survey about everyday writing practices and technologies. The demographics of students in these classes generally reflected the broader demographics of the school. Sixty-eight students from Ms. Kane's classes completed surveys, 38 of whom used she/her pronouns, 26 of whom used he/his pronouns, and four of whom used they/their pronouns. Approximately 60% of survey participants identified as Latinx,[1] 26% identified as Black, 7% identified as Middle Eastern, 3% identified as Native American, 2% identified as mixed race (Asian/Latinx), 1% identified as Asian, and 1% identified as white. None of the survey participants identified as English language learners. From Mr. Willis' classes, 63 students completed the survey, 26 of whom used she/her pronouns, 36 of whom used he/his pronouns, and one of whom used they/their pronouns. Approximately 51% of survey participants identified as Asian, 35% identified as white, 8% identified as mixed race (Black/Asian, Asian/white), 5% identified as Latinx, and 1% identified as Pacific Islander. None of the survey participants identified as English language learners.

Focus Group Participants

Focus group participants were selected from survey participants to discuss in further detail some of the ideas addressed in the survey with Jenni, who acted as moderator (Stewart & Shamdasani, 2014). Focus group composition was designed to represent a wide range of student experiences with and perspectives on technology and writing. Focus group selection was determined by survey responses, students' desire to participate, and the teachers' suggestions for focus groups (e.g., the teachers suggested potential focus group participants based on their knowledge of students' disciplinary interests, regular composition practices, etc.). Due to conflicts with standardized testing schedules at Inspiration, focus groups were not possible with Ms. Kane's students. Three focus groups, each with four students, were conducted with Mr. Willis' English 12 students. Table 1 provides an overview of the 12 focus group participants and their everyday digital and nondigital composition practices (self-reported in the survey and confirmed during focus groups).

TABLE 1
Student Focus Group Participants and their Everyday Composition Practices

Student, used pronouns, focus group #	Everyday digital and nondigital composition practices (listed in order reported)
Abigail (she/her), Group 1	Texting friends and family, emailing teachers, writing for school assignments
Des (they/their), Group 1	Creative writing, writing for school assignments
Nishant (him/his), Group 1	Texting friends and family, song writing, writing for school assignments
Sarah (she/her), Group 1	Texting friends and family, emailing, writing for school assignments
Carlos (he/his), Group 2	Texting friends and family, writing for school assignments
Warren (he/his), Group 2	Texting friends and family, emailing, writing for school assignments
Tracy (she/her), Group 2	Writing for school assignments, texting family and friends
Amara (she/her), Group 2	Writing for school assignments, emailing
Ishaan (he/his), Group 3	Creative writing, writing code, reminders to self
Kenny (he/his), Group 3	Text messages to family, emailing, writing for school assignments
Sonya (she/her), Group 3	Writing for school assignments, poetry
Matthew (he/his), Group 3	Texting friends and family, emailing, writing letters to friends

Data collection

Survey

The development of the online survey was guided by our research questions. Survey items were designed to gain knowledge about young people's varied writing practices in their everyday lives, their writing purposes, the writing tools they regularly employed, their familiarity with artificial intelligence in relation to writing, and their perspectives on AI and writing. The survey included "check all that apply"-format questions and open-ended questions (see Appendix A). Three open-ended questions about AI and writing were included so that participants could freely share their experiences and perspectives, using their own language and terminology (Braun et al., 2021).

Students in Ms. Kane's and Mr. Willis' classes were invited to complete the survey disseminated through Google Forms near the end of their respective units on social impacts of technology. Students were informed of a $25 gift card drawing incentive per class period, and they were sent one reminder email to complete the survey, which was open for two weeks. Given that the survey was disseminated at a very busy time (the end of the academic year for both Inspiration and Green Hills), the survey was designed to take under 10 minutes and to be easily accessible on mobile devices, as almost all students had their own smartphones. See Appendix A for survey items.

Focus Groups

Focus group interviews were used to encourage informal sharing of ideas among classmates who were familiar with

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one another and to avoid power imbalances that can emerge during one-on-one interviews, such as those between researchers and young people (Shaw et al., 2011). Focus group questions were phrased to elicit responses that could lead to group discussion of participants' everyday writing (digital and nondigital), their opinions and concerns about writing with AI, and their opinions about the social impacts of technologies such as AI (Krueger & Casey, 2014). See Appendix B for focus group protocol.

The three focus groups were held during the 90-minute class periods in which the students had Mr. Willis' English class. Focus groups were conducted in an unused classroom or lounge on the same floor in the building. At the beginning of a focus group, the researcher asked participants for permission to audio record their group's conversation; all participants granted their permission. Participants were also informed that they would be able to review their group's discussion transcription for accuracy. Each focus group lasted between 25 and 38 minutes. Participants were sent a copy of their group's transcription within 2 weeks of their focus group and were invited to alert the researcher to any errors or misrepresentations of their thinking. Participants did not report errors or misrepresentation.

Analysis

Data sources analyzed in this study include survey responses (131 completed surveys) and three student focus groups (12 students total). Both authors first reviewed all survey responses in Google Forms, which provided an initial, broad snapshot of young people's varied writing practices, their writing purposes, the writing tools they regularly employed, their familiarity with artificial intelligence in relation to writing, and their perspectives on AI and writing. Given our interest in young people's uses and understandings of AI for writing (or, to echo Pea and Cole, their doing and theorizing related to writing), we decided to focus analysis on the survey's open-ended response items related to AI, which asked participants to share their reasons for using or deciding not to use AI ("Whether you use AI for writing or not, what are your reasons for using or not using it?"), their perceptions of significant issues associated with AI and writing ("What do you see as some of the big issues around AI and writing that you or people your age are thinking about?"), and their developing critical understandings of the impact of AI writing technologies ("This year, you've done a lot of thinking about how technology impacts our lives. How do any of the ideas you've learned connect with your thinking about using AI for writing?"). Responses to these open-ended items were copied to Word documents and uploaded to the web version of qualitative data analysis program ATLAS.ti for coding. Recordings of the focus groups' discussions of AI and writing were transcribed by Rev.com and uploaded to ATLAS.ti for coding.

We adopted a collaborative approach to the analysis of the responses to the open-ended items to enforce the systematicity, clarity, and reliability of the coding process (Cornish et al., 2014). As a first step, we used first cycle coding methods (Saldaña, 2012) to apply attribute codes to different data sources (e.g., survey or focus group and associated teacher). We then independently read through Mr. Willis' survey responses and coded students' responses with deductive descriptive "topic codes" drawn from the research questions (*AI writing use*, *AI writing considerations*, *AI critical perspectives*). We added inductive subcodes to these descriptive codes to characterize the types of uses, issues, and understandings reported (Saldaña, 2012). After completing this first round of independent coding, we met to review similarities and variations in our coding results. We discussed our subcode definitions and reached consensus around our subcodes for each of the three topic codes related to our research questions (see Table 2 for topic codes, subcodes, subcode definitions, and data examples of subcodes).

With this initial codebook, we then coded Ms. Kane's open-ended survey responses and the focus groups' transcriptions to test the viability of our codes across the dataset. Independently, we found the topic codes and subcodes developed were applicable to Ms. Kane's students' responses and the focus group discussions. We also found that some students' responses aligned with more than one code. In those instances, we used simultaneous coding so that multiple applicable codes were assigned to a datum. After coding the entire dataset, we met to review our independent coding results and found we agreed in our coding, with a few variations in our simultaneous coding that were resolved through further discussion.

Once we completed our first cycle coding, we engaged second cycle coding methods, which are strategies that assist with "classifying, prioritizing, integrating, synthesizing, abstracting, conceptualizing, and theory building" (Saldaña, 2012, p. 58). During second cycle coding, we drew on cultural-historical approaches to writing and the construct of young people as philosophers of technology to guide our analysis. First, we created documents with coded excerpts related to each research question (i.e., youths' AI writing practices, key AI considerations, and critical perspectives). We then wrote individual memos about possible relationships between different codes, as well as how these relationships informed the notion of using *and* theorizing about technology (philosophizing) and the historically situated practices and contradictions that young people identified. We compared memos through collaborative discussion, during which we collapsed codes and noted predominating patterns corresponding to each research question. Through that process, we identified three patterns that we used to organize our findings section: (1) young people were deliberative about how they were using AI technologies (even when they

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TABLE 2
Topic Codes, Subcodes, and Data Examples

Topic code	Subcodes & definitions	Data examples
AI writing use	*Task*: Student describes use of AI writing tools to "offload" discrete tasks	"I use Grammarly for grammar and spell check"
	Explore: Student describes use of AI writing tools for entertainment or "non- serious" exploration (often in opposition to using it for school-related tasks)	"[M]ostly for non-school stuff that isn't actually important. I only really use it when I am messing around and not doing things that are serious"
	Supplement: Student describes use of AI writing tools to enhance or supplement an activity (e.g., support the thinking process)	"Use it for outlining since it can help get people started in writing when you don't know what the first step is."
	Non-use: Student describes not using AI writing tools for different reasons: they are against using AI; they do not want to try it; they tried it once and decided not to use it again; they do not have exposure to AI; they forget it is available	"my reason to not use it is because i would like to be able to learn about what im doing and not just get the answer given to by ai"
AI writing considerations	*Creativity*: Student describes an issue that relates AI use to creativity (usually in terms of aiding human creativity or threatening it)	"Creativity is dying out with the rise of AI"
	Authenticity: Student describes an issue that relates AI use to issues of "original" and "real" authorship	"I think one of the most prominent issues would be that AI can now copy and almost impersonate people through writing"
	Labor: Student describes an issue that relates AI use to work (toil) delegation, human effort spent or saved (energy and time), or human employment	"It does many tasks cheaply and quickly, eliminating jobs"
	Risks/harms: Student describes an issue that relates AI use to potential harms and risks (e.g., bias, misinformation, ease of academic dishonesty, loss of "human skills," inaccuracy of AI-generated information, impact on social futures)	"The data that AI is trained on is potentially biased leading to a biased AI. If writing in the future is created with AI then it will likely reflect the biases of society before AI that led to a skewed data set. It is important to make sure that AI is trained on data that does not reflect the power structures in our society so that it does not perpetuate them in the future"
AI critical perspectives	*Ethics*: Student describes issues of social and political significance and/or transformation in relation to AI writing technologies	"I see people talk about how some people who are in school to become doctors and nurses are using AI to help them cheat. These are the same ones who are going to actually be helping real people and if they're cheating now how do we know if they're actually good at their jobs"
	Future impact: Student describes imagined future effects or impacts in their discussion of AI impacts	"People are becoming too dependent on it as a resource and rely on it heavily to function in everyday life. I'm afraid that years down the line, society will see a massive decrease in scholars and graduates of school because of the abundance of AI assistants doing the work for them"
	Balanced perspectives: Student describes the need to develop critical, ethical understandings of technology in nuanced ways	"I feel like when we think about technology, we mostly talk about it negatively, which is you can make a lot of arguments that it is negative. There's a lot of ways it can be harmful, but at the same time, if you're trying to especially teach kids in high school more about it and how to stay safe online and stuff like that, you have to talk about negatives and positives because then it just comes off as like, oh, the teachers are just telling us that the internet is bad and technology is bad and AI is bad, but the world is changing literally every single day, so we can't control it. We just have to not be biased in what we're talking about and just have more perspectives, positives and negatives, and I feel like it's more impactful that way"
	School support: Student describes school-related supports to develop critical perspectives about AI	"The ideas we learned changed how I think about AI. How is our data used, how can the data it collects harm certain groups, how can we mitigate the harms of AI"

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were choosing whether or not to use them); (2) young people contemplated philosophical questions about the human dimensions of writing (and the boundaries between human/machine when writing with AI); and (3) young people raised ethical questions to help them grapple with an uncertain future with AI technologies.

Findings

The majority of students (91%) reported that they had used AI writing tools on more than one occasion. They reported using generative AI for diverse purposes in and out of school, including to accomplish routine organizational and information tasks, to entertain themselves through experimenting with AI technologies, and to catalyze their thinking and writing processes. In the first section below, we describe how these varied, layered purposes were deliberative, with young people deciding whether and how to use generative AI as one of many available composition tools that can mediate learning and life opportunities. We then explore how young people contemplated philosophical questions about what makes writing "human" as they grappled with what counts as "authentic" and "real" when the labor of making meaning is shared with machines. Finally, we discuss how students actively grappled with critical issues surrounding generative AI, both within and beyond the classroom. This involved placing ethical dilemmas at the forefront and facing the uncertainties of a future living alongside intelligent machines. These findings counter monolithic constructions of youth as unconcerned, uncritical users of technology and highlight the important role schools can play in supporting young people's sociotechnical inquiries. In our report of findings, students' responses are followed by the initial of the classroom teacher's name and the number assigned to a particular survey response.

Youth Contemplating Their Everyday Uses of AI: Deliberative Decision-Making

Challenging the misconception that young people primarily use generative AI to bypass academic writing, our study revealed a rich tapestry of motivations for writing with AI among participants. Participants demonstrated careful consideration, noting specific and thoughtful justifications for their decisions about whether, how, and under what circumstances they turned to AI. Students reported that generative AI tools helped them manage specific, discrete tasks that they encountered over the course of a day, as one student elaborated: "Summarizing large documents, writing templates for emails, helping with structuring my coding assignments, explanations of difficult-to-understand economics ideas, chemistry questions, and understanding jokes" (W38). For several students, AI use provided an information shortcut that allowed them to "find information faster without having to look it up and sort through different browsers to find what I am looking for" (W58), which ultimately helped them "get information on a topic" (W10) or "learn more about a certain topic" (K44) much faster.

Focus group participants in Group 2 spoke at length about the use of AI to address everyday tasks, especially in relation to how generative AI streamlines routine academic tasks such as synthesizing information across multiple sources. As Sarah explained, "instead of looking for five sources to combine all of them to make one answer, you get one answer just from a few words." This topic was also raised by students in the first focus group, although one student, Nishant, reported that sometimes the need to use AI was driven not by expediency but by gaps in resources: "I did AP Physics online because we don't have a teacher in the school, so a lot of times with conceptual questions and stuff like that, I'll just ask the AI because I don't really have a teacher."

Students' recognition of AI-addressed needs related to knowledge or resource gaps also surfaced in survey and focus group participants' reported uses of AI for low-level composing tasks, especially grammar and spelling. Students stated that they used AI because they generally felt "unsure whether my grammar is correct or if a sentence is written properly" (K42). In some cases, AI was viewed as a necessary support related to prior learning experiences: "I was never formally taught grammar rules, so I use Grammarly a lot because I don't want to make mistakes" (W9). Most of the students who reported using AI for low-level composing tasks were cautious about overreliance on the tool because "using it too much makes me too dependent on the AI to check for grammar and spelling mistakes" (W11) or because using it for anything other than error detection "doesn't feel genuine" (K66). This worry related to overreliance on machines to replace human effort was a recurring idea intertwined with other issues, as we delve into further in the next findings section.

For some students, AI acted as a thought partner to supplement or support their composing. As one student described, using generative AI "acts like a catalyst for my thoughts" (W13), a viewpoint shared by others who similarly reported that AI offered "a spark to go off from" (K32) and "ideas to go off of for my writing" (W15). This notion that generative AI is a catalyst for thinking and writing, or a means of "[getting] ideas and new perspectives" (K16), appeared in over a quarter of the survey responses. Some students suggested that this could look like generating ideas, while other students described using AI to generate ideas to aid other dimensions of the composing process: "it can give me an outline or gameplan on how to write a paper" (W41). Participants in Focus Group 2 elaborated on this idea of AI-as-catalyst, with students clarifying that

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this particular use of AI functioned like other supplementary technologies that writers use to support the broader writing process. Carlos explained, "If I'm stuck on a writing prompt, I kind of put [the topic] in there and then I just write off what it said. I don't copy it or anything, just kind of get the ball rolling for me." Carlos' careful qualification of how he used AI as a supplement (i.e., he did not use AI to copy and take the "easy" path) was echoed in survey responses, as students made clear that using AI as a supplementary tool did not take away their voice or autonomy as writers, but actually served as a generative resource that would help them "to get the ball rolling."

More than half of the students reported that they were introduced to generative AI through experimenting with tools they tried "for fun," a phrase that recurred across the data (see Stornaiuolo et al., 2024, for more about this dimension of student use). Snapchat seemed to be a central site for this playful experimentation for several students: "I don't think I've used AI other than MyAI on Snapchat and just messing with it" (W53). Some students also shared that AI could alleviate boredom or loneliness, as suggested by this student: "Al on snapchat can also be used for just if your by yourself and you want someone to talk to" (K1). These exploratory uses of AI often revolved around students' curiosity to see what the tools could create ("I used snap AI and chat GPT to see what kind of answers they would generate and whether they were accurate or not" [W45]) and whether the tools fulfilled or exceeded expectations ("The reason I checked it out was because people didn't shut up about it" [K24]). However, students were very careful to distinguish their "messing around and not doing things that are serious" (W21) as "only as a joke and not for school" (W22). As we explore later in this paper, fears related to academic dishonesty may have prompted these qualifications, as evidenced by the frequent inclusion in survey responses (which participants knew would be read by adults) of phrases emphasizing independent work: "I prefer to do the work myself and use the skills I've learned" (W49).

Even though most students (91%) reported experimenting with generative AI, about 27% indicated that they did not use generative AI regularly or at all. A small number of students in this group were unfamiliar with AI technologies ("In all honesty I don't really know what AI is" [K28]) or did not know how or why they would use it ("i don't know how to use it" [K14]). However, the majority of the students engaged in what we called principled non-use, as they either refused or made a deliberate decision not to engage with it. These students expressed a variety of reasons that guided their intentional rejection of AI writing tools, with some stating that they prefer "my writing to be original" (K67) and "to be able to learn about what I'm doing and not just get the answer given to by ai" (K47). Only one young person who did not use AI and who identified as Black alluded to how uses of AI might be tied to broader racial, social, or socioeconomic factors: "what i see is that people like me doesn't use it a lot" (K2). Prevalent concerns related to originality and authenticity were often connected to issues of fairness and ethics. As one student said, "I have only used it once and it was just to try it out. I am against plagiarism and I don't feel using chatGPT or any other AI writing is ethical" (W26). Others echoed the sentiment that "it's not fair to write using AI" (W28) and that "it doesn't feel authentic to me and feels like cheating" (W54). Underlying many of these choices not to use AI was "non-confidence in coherency" (W40) of AI-generated writing, feeling "scared of plagiarism" (W3), and desiring to retain autonomy over one's own words and voice ("I get a lot more control when I write the material myself" [W17].). One student summed up these concerns, which are taken up more fully in the following section, when they said, "I don't use Ai for my writing because it counts off the creativity that an individual has and you can't really control how the writing is going to be" (K9). Concerns related to writers' autonomy and creativity, central to student perspectives about using AI for writing, proved to be a significant deterrent for some students, who opted not to engage with AI technologies altogether.

Youth Contemplating Issues of Creativity, Authenticity, and AI Use: The Importance of Being Human

One of the most striking findings in the data centered on young people's engagement with philosophical questions regarding the essence of human writing, particularly in relation to how human creativity and voice are being redefined in the context of machine-assisted composition. More than a quarter of survey respondents highlighted the creative affordances of writing with generative AI, as expressed by Ishaan during his focus group discussion: "One thing I like about AI is kind of the creativity it can possess, you know? You can just play around with it and can just have fun with it." In this comment, Ishaan reflected on how AI can become a creative tool if "you" (human writers) wish to explore the possibilities. Survey respondents and students in other focus groups similarly acknowledged how AI could support human creativity. Des, in Group 1, emphasized how writing for them was a creative process that was distinct from AI composition: "I'd rather just write authentic my type of writing or whatever.… I kind of like incorporating my own creativity in it and writing and stuff." Nishant, also in Group 1, concurred with Des' views on creativity as a uniquely human endeavor: "I think it [AI] just lacks originality and creativity that people have." These observations suggested that young people were willing to think with and about AI, but only to a certain extent, and not if they perceived AI would compromise their creative processes and/or their understandings of technology and human nature. As one

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student noted in their survey response, it was important to keep AI technologies "in their place": "[AI] should be used as a tool and nothing more. It's the only way to keep our creativity ours" (K38). By distinguishing and enclosing AI as a "tool," this student suggested, writers could protect their human creativity (collectively "ours"). These kinds of positionings of AI as other and writing as human were common throughout the data, indicating that students believed some level of vigilance is necessary to prevent non-human tools from encroaching on human territory (in this case, the human technology of writing).

Anxieties about AI as a threat to human creativity surfaced repeatedly in students' responses, revealing a contemplative orientation to philosophical questions about the shifting nature of writing with AI technologies. Youth expressed concerns over how AI could diminish humans' ability to create and be creative (which they understood as the emotions, perspectives, and imaginative capabilities that make people unique), and they grappled with what they should do in response. Many students characterized using AI as a danger to their own creativity. As one participant shared, "I don't use AI for writing because I enjoy being creative with my writing. Writing is a way I can express my thoughts and feelings and if I have an AI do it for me, it takes that away" (W57). Several participants also stated that they protected themselves from what they perceived as untrustworthy AI writing tools by avoiding them altogether and instead relying on their own writing abilities: "I do not use AI for writing because I would prefer if my work is generated from myself and I don't entirely trust neither am I comfortable with what AI produces" (W7). Other students who decided against AI expressed that they felt unsure how to assess acceptable AI use for schoolwork and therefore avoided potential accusations of academic dishonesty by strengthening their creative capacities as writers: "I just found that I would rather use my brain and train my creativity rather than rely on AI. I also fear that using AI for schoolwork will get me in trouble. I don't know whether or not its original enough to not be accused of plagiarism" (W32). Uncertainty about what constituted plagiarism and worry about consequences meted out by adults and institutions who hold the power to define plagiarism, was a significant concern, particularly for youth who already experienced marginalization based on their identities.

Quite a bit of conversation revolved around the threats of AI to creative enterprise more broadly, whether that involved professional writers or artists, concerns over "deepfakes," or worries about how ChatGPT dataset was trained on people's texts without consent. As Amara discussed the Hollywood writers' strike in Focus Group 2, she summed up the group's concerns that "a lot of creative minds are being replaced with AI." These concerns over ceding creative jobs to AI seemed rooted in fears that AI might compromise what it means to be human, as one survey respondent detailed:

> I think there's always been a fear of robots taking over the world, through "stealing jobs," or through a robot uprising. These anxieties are even stronger now that language models like ChatGPT are increasing in popularity. Personally I don't think that there is a threat of AI taking any creative job like writing or painting because what AI can create seems limited in originality and is usually pulled from databases of human-created art and writing. I think a big part of what makes art and writing enjoyable is the human experience and emotions we feel, which is something that AI does not yet have the capability to experience (K64).

This student suggested that collective fears of AI mirror growing anxieties about humans' relations to machines that "are increasing in popularity." The student noted that these social anxieties loom large in the collective consciousness, even though they believe that AI is incapable of knowing and representing "the human experience and emotions we feel" as writers, which are central to the experience (and pleasure) of writing, suggesting their active negotiation of different perspectives on generative AI (in this case, their own juxtaposed with popular narratives concerning machine take-over).

Students' concerns about human creativity were often connected to contemplations of the nature of authenticity. In their survey responses and group discussions, students examined the boundaries of real and artificial in relation to humans and machines, often expressing confusion as they shared their emergent sensemaking. For example, students expressed concerns about how to demarcate, and protect, their own words (and meanings): "it [AI] might help with school work but not too much because its not your words" (K15). The dilemma of "how much AI is too much AI?" surfaced in students' wonderings about how much AI they could use before it compromised their writerly voice: "I think it should really be used as a tool if necessary but to be honest don't rely on it too much or else it'd be something else's words not yours" (K23). As one student noted, what one can do with "AI technology" is "separate from authentic creativity that people will need to use in everyday life" (W7). Comparisons of human and machine writing were threaded throughout the data, with machines usually deemed incapable of conveying qualities students associated with "real" composition: "AI generation is pretty soulless even for my standards and it doesn't use the same dramatic diction I use" (K29). Indeed, we noted that this idea of "soulful" human writing appeared in a number of responses, always as a critique of AI- mediated composition: "I don't like AI especially AI art or AI music I think it takes away the actual soul from those pieces" (K37). Observations of the perceived soulful nature of human writing were often juxtaposed with descriptions of "empty," "bland," and "lifeless" prose generated by AI. As one survey participant explained: "The reason why I don't use it for writing is because I can't put any emotion into my writing. I feel like when I write essays I am always trying to put my all into it. So if an AI would write for me and I would read it I know

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my flavor and creative writing isn't in it. It's all just a lie" (K39). When machines composed, students agreed that the products were "a lie," or not real (i.e., human, emotion-filled, soulful) writing. The greater danger of continuing down the AI path was that writing itself would become less of a (genuine) human activity: "Over time everything might be written by AI and articles won't be as genuine and or human connected" (W60).

Associated with these concerns about authenticity and creativity were ethical wonderings about the labor involved in creating meaning with AI and how to assess uses of AI that were congruent with students' understandings of "honest" writing and "cheating" (i.e., the offloading of intellectual labor to machines). Some students decided to use generative AI only for error detection, since other uses seemed more aligned with cheating: "Like I said I only use the spell check feature that's built into the computer, I don't know if that counts as the ai that is being referred to in this survey but I only choose to use this one because the others seem like cheating" (K5). For many of the young people, familiar and institutionally accepted tools such as spell check and grammar check seemed the only ethical applications of AI for writing. Others found that using AI for revision was useful but still acknowledged that as a result, the writing would not be entirely one's own: "I guess is to put something that you already wrote and it will give you the answer that you wrote but in the ai way that you ask it to do then after that I guess you can rephrase it so it can still become your work most of the time, not 100%" (K7).

Almost every student in the survey and focus groups expressed concern regarding AI use and plagiarism, even when they were doing much of the labor (i.e., idea development through writing) themselves. These concerns seemed rooted in recognition of narratives constructed by adults about AI and youth. One student suggested that adults were more interested in perpetuating stereotypes of youth using AI for cheating than in understanding what youth were actually doing with AI: "That AI is used for cheating is mostly believed by older people when most of my peers use the Ai as tools to help our writing" (K25). As this student suggested, young people found themselves making decisions about using AI to "help our writing" in ways that adults may not understand or may construe as plagiarism. Identifying how to share intellectual and composing labor with machines and what uses might be deemed by adults as assistance versus cheating was a significant dilemma raised by youth across the datasets.

Youth Contemplating Critical AI Issues: The Importance of Ethics in Human/Machine Futures

One pattern we noted was that students tended to approach generative AI with a critical lens. This critical perspective might be linked, in part, to Ms. Kane's and Mr. Willis' sociotechnical teaching approaches, but it also likely reflects wider conversations about AI in society (e.g., Spector, 2023). Ms. Kane and Mr. Willis had facilitated critical conversations about technology throughout the year, and we noticed students connected classroom discussions about the harms and risks of technologies to their perspectives on and practices with AI in surveys and focus groups. For example, one student highlighted the bias in AI that might get inscribed into their own writing inadvertently: "Probably [my biggest concern is] the inherent bias in AI. Because using AI for writing can lead to you having bias in your own writing without that intention" (K58). This student seemed to suggest that a major concern with using AI is that the bias and discrimination coded into algorithms (Benjamin, 2019) can be folded unwittingly into a writer's voice. Other students made similar connections between writing with AI and the reproduction of harmful social hierarchies, noting that "[i]f writing in the future is created with AI, then it will likely reflect the bias of society before AI that led to a skewed data set" (W62). For these participants, writing with AI "doesn't seem like a good idea" because the produced writing "would likely not be the best work I can produce because of the bias in algorithms" (W60).

At the heart of students' critical stance toward generative AI lay ethical concerns, particularly around "fairness." These concerns stemmed from uncertainties and anxieties about whose voices and perspectives are represented and demarcated when collaborating with machines. One student described how the training data scraped other people's writing without attribution: "There is also the issue of plagiarism because AI works in a way that it learns from previous existing examples so it could be argued that using AI steals the work of others" (W62). As this student implied, because AI writing recommendations are always "scraped" from unattributed sources, there is no way to avoid plagiarism. Other students noted that writing with AI also made intentional acts of plagiarism easier to commit. A common refrain was that using AI was "the easy way out" (W8), or a way of "taking advantage" of AI to cheat: "I think people are taking advantage of the new technology and using it to cheat their way through assignments, work opportunities, and even relationships when they don't know what to say" (W26). These concerns about cheating often revolved around ethical concerns about unfairness related to effort made (labor): "I think a lot of the issues are with doing your own work. AI gives you a way out of writing, which I think is unfair" (W28). The issue of unfairly "getting out of writing," as a form of shirking intellectual responsibility, was often tied to disadvantaging those who did not use the technologies: "Using AI to completely write your essays is not fair for others who write their essays honestly" (W11). One kind of concern about fairness that came up repeatedly was that people

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would become over reliant on machines, with significant consequences: "Many people that I know write their school work and college essays with AI and that diminishes the work of people writing completely on their own and prevents students from learning how to write" (W62). This worry about "the way we rely on AI to do things" (K58) instead of learning about writing seemed rooted in broader concerns over labor and meaning.

Students' ethical concerns about bias, cheating, and fairness were not only tied to immediately relevant issues but also to uncertainties about a future in which AI is poised to have greater impact over time. The young people in our study wanted to know "How will AI change our schooling? How will it change our future job processes?" (W12). In Focus Group 3, Sonya related future AI developments and related problems to issues of authenticity: "[AI] is developing pretty quickly. So I feel like if it just keeps going at this rate, I don't know, it'd kind of be a little concerning because then you can't tell what's real or what's not." Students' ponderings about AI and the future often lingered on "scary" uncertainties associated with AI use and the equally disturbing potential consequences: "Using AI for writing is scary because we know everyone has a digital footprint. If you get caught for cheating at school, it would go on your record and it's not worth it" (W3). This student knew that data were being compiled about them and that any negative repercussions, such as being accused of cheating, would follow them into the future. Other students linked these concerns about the future to AI programmers' design decisions, particularly in relation to how power shapes technology development: "The data that AI is trained on is potentially biased leading to a biased AI. If writing in the future is created with AI then it will likely reflect the biases of society before AI that led to a skewed data set. It is important to make sure that AI is trained on data that does not reflect the power structures in our society so that it does not perpetuate them in the future" (W62). Some students even questioned whether writing, regarded by many participants as an essential "human" activity, would continue to be taught in schools:

> I'm sure some people are wondering why we learn how to write in the first place. It asks all of us the question, "If the robot can (eventually) write better than us and faster than us, why do we need to learn how to write?" One might argue that it is necessary for you to survive in society, but in a future with technological advancements unimaginable to us, I think there might be a point where writing will be obsolete (W17).

In this student's comments and in others, questions about the obsolescence of writing seem to also be questions about the obsolescence of human activity in other domains. Students wondered if people "will no longer be hired when AI is such a convenient, efficient, and cost-effective product" and if, as a result, people will "no longer [be] able to exercise their creativity and future generations will decline" (K32). The many worried references to the future suggested that young people were taking seriously the potential consequences of AI technologies, tracing a fraught line between their current decisions and practices and the unknowable future.

Focus group discussions suggested a concerning gap in schools: critical and ethical dialogues regarding the intertwined futures of humans and machines were largely absent from students' school experiences. Even though they had experienced Ms. Kane's and Mr. Willis' sociotechnical curricular designs, students of both teachers reported that an overall lack of learning opportunities in school contributed to their uncertainties about the social impacts of technology. Students observed that even if technological impacts were discussed, teachers were likely to vilify technology. Sonya (Focus Group 3), for example, raised the issue that "when we think about technology [in school], we mostly talk about it negatively." This approach, she argued, failed to prepare students for the complexities of constantly evolving digital (and nondigital) environments:

> There's a lot of ways it can be harmful, but at the same time, if you're trying to especially teach kids in high school more about it and how to stay safe online and stuff like that, you have to talk about negatives and positives because then it just comes off as like, oh, the teachers are just telling us that the internet is bad and technology is bad and AI is bad, but the world is changing literally every single day, so we can't control it.

In Sonya's account, the importance of presenting a balanced view of AI was to help students know how to act when they were not able to "control" the changing nature of the technological landscape; negative prohibitions against the technologies were ineffective. Kenny concurred with Sonya, adding that "to be conscious of how you use technology in your everyday life, [knowing the] benefits and harms… [allows you] to just use technology in good ways to help the world." Kenny's observation suggests that awareness of balanced perspectives on technology use can help individuals develop agency for informed action (as did the students in Lee et al.'s, 2022 study; the need for such balance was also echoed by high school researchers in Singer, 2023). The other students in Focus Group 3 agreed with Kenny and Sonya's views, with Matthew describing his desire for schools to engage students in understanding the complexity of sociotechnical issues:

> So what it means to be a critical user of technology, like I said earlier, is to be able to use it efficiently and effectively at the same time. And to mitigate the harms and know how it impacts your everyday life. Know what it could do to help your life and know how it harms your life. But that's about it. Just being able to use it efficiently and effectively and mitigating the harms.

For Matthew, mitigating harms and knowing the impacts on one's everyday life were key dimensions of

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engaging AI in critical ways. Students across focus groups and surveys echoed these desires to learn about balanced, complex perspectives about AI that recognized the benefits and fun of "messing around" even as one learned about ethical responsibilities of fairness and harm mitigation.

Discussion

The present study drew on student survey and student focus group data from English language arts classes in two culturally and linguistically diverse high schools to answer the following questions: (1) How are young people using AI in their everyday lives, if at all?; (2) What do young people identify as key considerations related to AI-mediated writing?; and (3) What ethical and critical considerations, if any, inform young people's sensemaking of and practices with AI? Countering narrow constructions of youth as eager adopters of technologies that will allow them to "offload" academic writing and thinking, the youth in our study showed complexity in their AI writing practices and in their contemplation of their practices. Findings from survey and focus group discussions suggested that young people use AI deliberately and thoughtfully for varied writing purposes in and out of school, including to accomplish routine organizational and information tasks, to entertain themselves by creatively composing with AI technologies, and to catalyze their thinking and writing processes. Findings also suggested that participants regularly navigate ethical and critical dimensions of AI use, and that these navigations draw them into contemplating what counts as "real" and "authentic" writing and, relatedly, what it means to be human through and with advancing technologies. Such philosophizing, situated in their lived experiences, offers valuable insights for literacy educators and researchers, specifically around how generative AI may co-opt, suppress, and/or overwrite young people's voices in ways that are not always immediately apparent to adults.

Many participants in our study expressed uneasiness with using AI for writing due to uncertainties surrounding representation and ownership of ideas and voices. Even students who noted potential affordances of AI use also raised concerns, particularly around the potential for biases, discrimination, and unfairness. There was not a single student who voiced wholehearted approval of generative AI. This uneasiness seemed related to their perceptions of what counts as "real" writing activity and how "real" writing develops. Despite being well-versed in various digitally mediated forms of writing (e.g., social media posts, video design, music composition), the students voiced suspicions that AI writing is different from "real" writing and a threat to (real) writers' creative abilities. As Nishant observed in his focus group conversation, AI writing "just lacks originality and creativity that people have." These concerns, seemingly grounded in students' perceptions of writing as an individualistic activity that requires a person's unique creative capacities, suggest that young people may be recognizing that generative AI introduces not only new ways of writing but also new models of literacy altogether. As Robinson (2022a) has argued, rapid advancements in natural language generation platforms may reflect the emergence of a new autonomous model of literacy that reshapes writing by redefining/redistributing writerly agency and meaning-making across sociotechnical entanglements. Young people's worries about the potential loss of "real" writing through AI use suggested that they valued rather than spurned the hard work, emotions, sparks of creativity, setbacks, and laborious meaning-making that they associated with writing.

Our study participants' contemplations of "real" writing versus machine writing also created space for many of them to examine what it means to be human. Through their theorizations of what AI is *not* (original, soulful, capable of emotion, critical, accurate, reliable, etc.), these young people also constructed what they believe humans to be. Comparisons of human and machine writing were threaded throughout the data, with machines usually deemed incapable of replicating qualities the students associated with humans. While some students' worries seemed tied to conflations of generative AI and general AI (the type of artificial intelligence that matches or exceeds that of humans), even expressed fears of an impending machine "take over" touched on students' understandings of human capacities to explore, develop and maintain relationships, and create—with or without AI writing technologies and "technological advancements unimaginable to us" (W17). Several students' observations also evidenced their comprehension of human biases and the risk of AI inheriting them, thus highlighting their awareness of complex sociotechnical entanglements. Young people's philosophizing of human-machine possibilities, tensions, and futures in the context of their own composing lives contributes nuanced perspectives in which mediational and sociotechnical dimensions converge, and in ways that might inform how literacy educators approach writing. Their concerns about creativity and authenticity in writing, along with their fear of becoming reliant on machines, highlight how much they value writing as a human endeavor, as well as their worry that AI may eliminate opportunities for meaningful writing. This apprehension seems to stem from the idea of AI as a crutch, and a fear of losing the ability to write well enough to stay in control of both the technology and their futures. Contrary to some assumptions about young people and AI, our participants were not looking to delegate thinking and writing to machines. Instead, they expressed a strong desire to engage in "real" writing that they find valuable, and perhaps especially writing that allows for aesthetic expression.

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In contrast to earlier digital writing work in which scholars found that young composers intertwined aesthetic and ethical dimensions in digitally mediated compositions (e.g., Hull & Stornaiuolo, 2014), we noticed that youth in our study tended to separate ethical and aesthetic considerations in their discussions of AI writing and "real" writing. Ethical concerns were associated with writing with generative AI, while the aesthetic dimensions of writing were related to "real" writing done by people. Although a few students commented on what might be created with AI (e.g., literature, music, film), only one expressed appreciation for new possible art forms that might emerge from AI-human collaborations. In general, aesthetic considerations (beauty, pleasure, play, depth of emotion, expression) were linked to human activity. This separation of aesthetics and ethics suggests a need to explore how educators can support writers in engaging both of these important dimensions as new writing practices and forms continue to develop.

While we view young people as philosophers of technology even without critical lenses developed through designed curricula (such as the kind our study participants experienced in Ms. Kane's and Mr. Willis' classes), we also recognize that those academic learning experiences provided opportunities, still uncommon in most English classes, to theorize about how they compose with AI, or choose not to compose with AI through different texts and activities (Pea & Cole, 2019), and how their composing practices are always situated locally and globally in wider entanglements of people, histories, material and psychological tools, sociopolitical dynamics, skills, knowledge, bodily coordination, and spatial and temporal circumstances. As focus group participants noted, there is value in adding these opportunities to gain more "balanced" approaches to emerging and encroaching human-machine configurations.

Conclusion

We are left bolstered by the depth of philosophical inquiry displayed by young people in our study. They delved into fundamental questions about the nature of writing in the age of generative AI, exploring what makes writing uniquely human and grappling with ethical choices about integrating these emerging technologies into their writing lives. While these young people wrestled with profound questions about writing with machines, their multifaceted responses revealed a nuanced understanding that there are no simple answers to the dilemmas and uncertainties they considered. The complexity of their perspectives reinforces the importance of including young people as central stakeholders who can enrich conversations about AI and in educational research more broadly (Lyiscott et al., 2020). Indeed, we were acutely aware that our position as adults and the power dynamic may have limited how much young people shared with us. For example, while concerns about unethical uses of AI were raised, no participants reported engaging in such practices themselves. We wonder what more may have been learned had this study included youth researchers or participatory methods.

Discussions about AI ethics within educational research, practice, and policy with and about youth must also put access and equity at the center, as we ask: who is learning about these ethical issues with AI, where, and under what circumstances? While we did not engage in a comparative analysis of the two schools in this paper, we observed some noteworthy differences that warrant further investigation. For example, while no students reported being unfamiliar with AI at Green Hills, the more affluent school, there were five students at less financially resourced Inspiration who reported being unfamiliar with generative AI or unsure about how to use it. Furthermore, students at Inspiration, a school with a predominantly Black and Latinx student body, exhibited greater apprehension about potential accusations of cheating and expressed greater uncertainty about how AI might affect them. We find these trends troubling as they align with existing research on how emerging technologies are used for racial discrimination and surveillance (Benjamin, 2019), particularly for BIPOC youth writers (de Roock, 2021). Given that schools serving low-income communities often utilize technologies in ways that exacerbate inequities (Reich & Ito, 2017), we argue here for the importance of designing learning opportunities in schools that engage young people in sustained and situated inquiries about AI use (and about the social impacts of technology more broadly) to help address these potential disparities.

While the youth in our study were engaged as philosophers of technology as they "messed around" with and learned about AI in their everyday practice, they also pointed to the important role educators and schools could play in supporting and deepening their philosophical endeavors. One implication from our study is that schools can and should do far more to support students in thinking critically about the role of these technologies in their writing lives – not in a punitive or simplistic way (e.g., don't use AI) but through deep, sustained, and balanced inquiry. Such an approach would honor the playfulness and sense of fun that young people bring to their practices with new technologies (as our participants reported), and also help them thoughtfully and critically engage with and take informed actions to address the everyday ethical dilemmas that surface in those practices, from considering how their writing practices connect to broader social, political, economic, and historical forces, to how they can use new technologies in ways that mitigate harms. Such an approach could open new opportunities for educators and students to think together about what it means to write and be a writer when one's (creative, authentic) voice can be

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algorithmically distributed, and what it means to be human within our rapidly evolving sociotechnical systems.

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Conflict of interest statement

The authors certify that there are no conflicts of interest.

ENDNOTE

[1] We use Latinx to recognize nonbinary gender identifications among individuals who are of or relate to Latin American origin or descent.

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APPENDIX A

Student Survey Items, "Writing in the Digital Age"

The purpose of this survey is to learn about the different ways people your age use writing in their everyday lives, with and without digital technologies. Your ideas are important, and they will help teachers and researchers understand issues about writing that matter to young people right now. Thank you for sharing your thoughts!

One name will be drawn from each class period for a $25 online gift card to Amazon.

1. "Writing" includes but isn't limited to words. For example, all of this counts as writing: typing essays, creating/editing videos online, handwriting notes, texting, emailing, writing in a journal, using technology to help generate writing, composing songs/lyrics, posting in online communities, collaborating on fan fiction, etc. What kinds of writing do you do the most in your eve ryday life?
2. What motivates you to write? Check your top 3 reasons for writing from the list below.
   - School
   - Creativity
   - Communication
   - Personal (to remember, process, or comfort)
   - Sharing (to see how others respond to you)
   - Other
   - Add option
3. What writing tools do you use most often when writing? Check anything that you use multiple times per week. You can add others as needed.
   - Paper and pen/pencil
   - Computer
   - Tablet
   - Google products (Docs, Sheets, Slides, etc.)
   - Microsoft products (Word, Excel, Power Point, etc.)
   - Video or music editing software
   - Phone
   - Other
   - Add option
4. Have you ever used artificial intelligence (AI) for any kind of writing (social media, school work, etc.)? Check any tools that you have used or tried:
   - My AI on Snapchat
   - ChatGPT
   - Sudowrite
   - Grammarly
   - Bard
   - Quillbot
   - Other
5. If you use or tried AI for writing, what do/did you use it for? Check the box that applies:
   - School-related writing
   - Non-school writing
   - Both school and non-school writing
6. If you don't use AI for writing, do you know people who do?
   - Yes
   - No
7. Whether you use AI for writing or not, what are your reasons for using or not using it? [short answer text]
8. Who do you see as some of the biggest issues around AI and writing that you or people your age are thinking about? [short answer text]

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9. This year, you've done a lot of thinking about how technology impacts our lives. How do any of the ideas you've learned about connect with your thinking about using AI for writing? [short answer text]
10. Thanks for sharing how you are writing now and the issues you are thinking about. Here is a space for anything else you want to add that we didn't ask! [short answer text]

APPENDIX B

Student Focus Group Prompts

1. What are the different kinds of writing you do in your everyday life and for what purpose?
2. Something that came up in many survey responses was being worried about writing with AI. Let's talk more about that. What worries you?
3. What are issues related to writing with AI that you think people need to consider?
4. What do you think it means to be a critical user of AI technologies such as the ones you use to write?
5. Thinking about what you learned with Ms. Kane/Mr. Willis this semester, what comes to mind when you think about technology's impact on society?

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